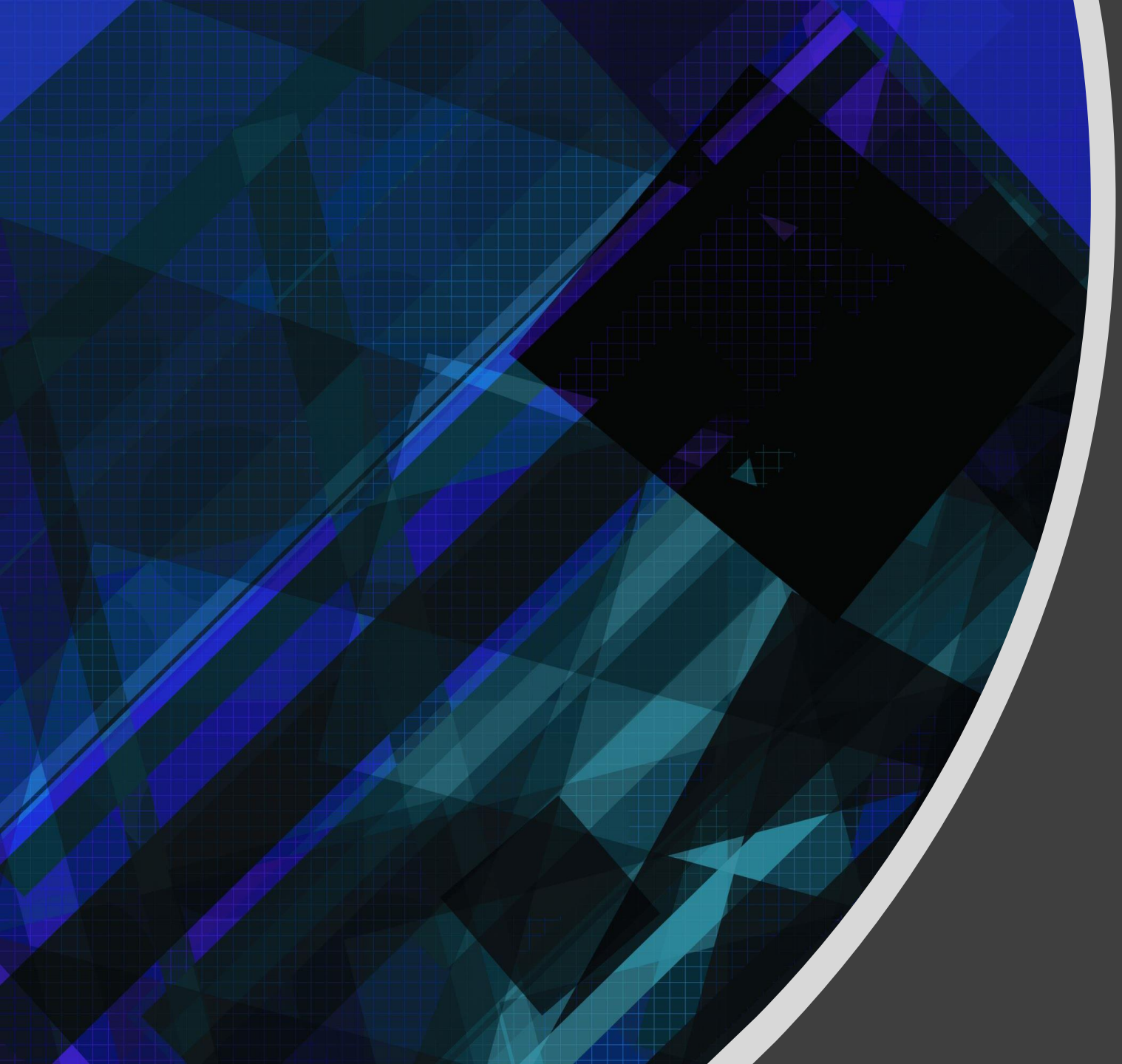


# Gestures

# Prayer

- Lord God we thank you for your love and countless blessings.
- Please Forgive our sins and short comings.
- Bless our teacher with knowledge, wisdom , and skills.
- Help us to grow , learn , acquire skills and build our characters to become a better version of ourselves.
- Guide us to live according to your will so that we will live a fruitful , blessed and successful life.
- We give your glory and honor , In Jesus name , Amen.

"Clothe yourselves with compassion, kindness, humility, gentleness and patience." - Colossians 3:12

- When have you been the benefactor of someone's kindness? How can you show kindness to another?

- Jesus, thank You for showing me overflowing, unlimited kindness. Help me to find joy in doing kind acts for others. Good morning. A blessed Sunday.

# Review and Feedback :

# Flutter Architecture

- The Flutter architecture mainly comprises of four components.

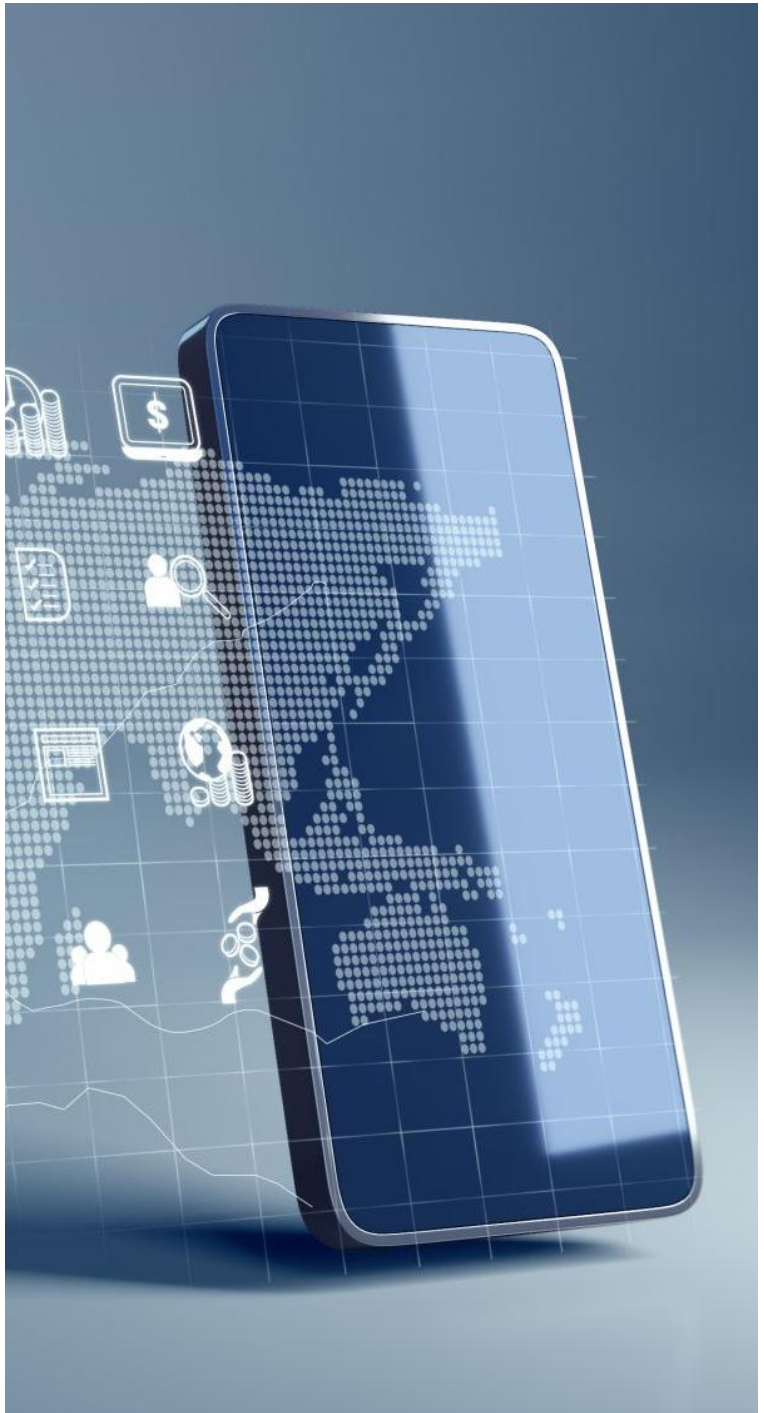
1. Flutter Engine
2. Foundation Library
3. Widgets
4. Design Specific Widgets

# Widget types:

- **statefulWidget**
- **statelessWidget**

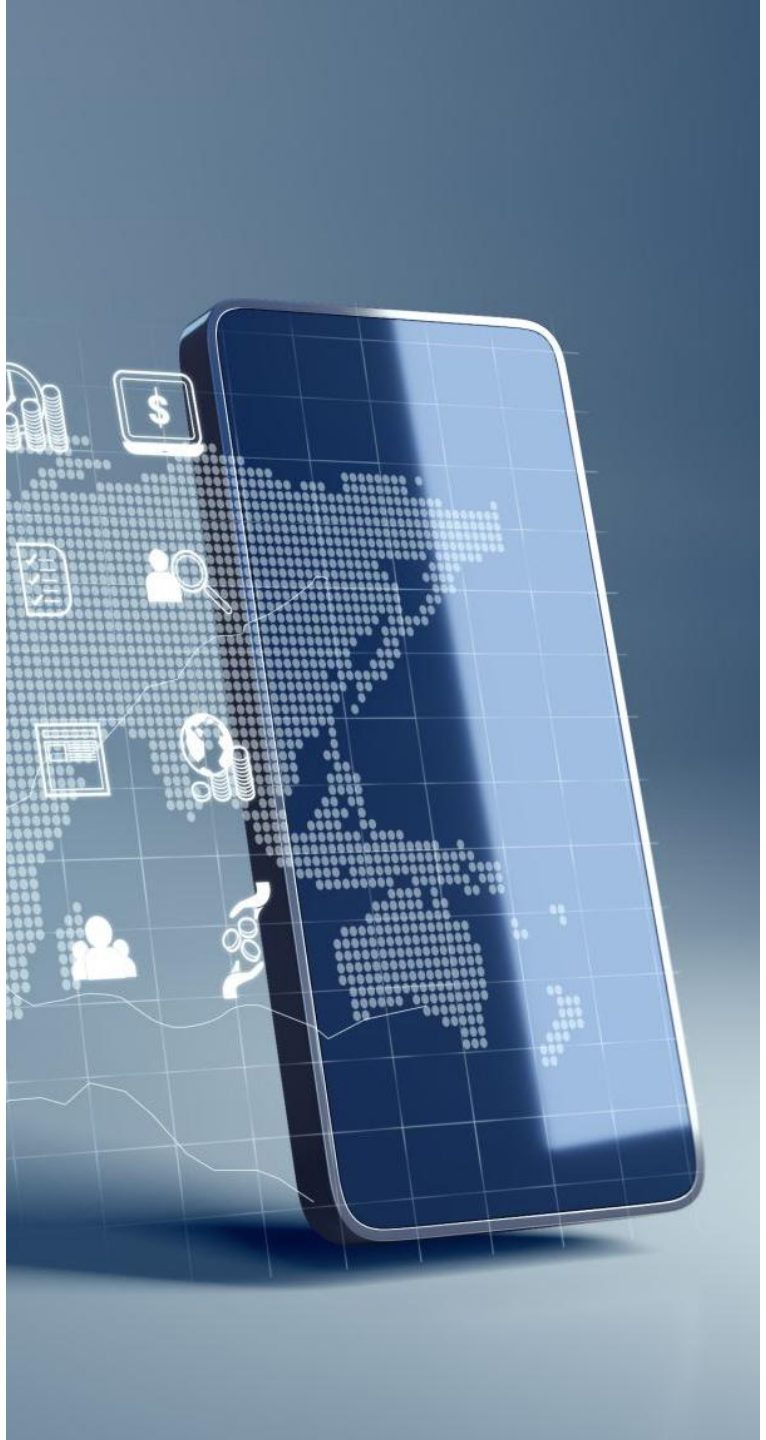
# Flutter Layouts categories

- Single-child layout widgets
- Multi-child layout widgets
- Sliver widgets



# Standard widgets

- _____: Adds padding, margins, borders, background color, or other decorations to a widget.
- GridView: Lays widgets out as a scrollable grid.
- ListView: Lays widgets out as a scrollable list.
- Stack: Overlaps a widget on top of another.



# Standard widgets

Container ListView Stack GridView

- _____: Adds padding, margins, borders, background color, or other decorations to a widget.
- _____: Lays widgets out as a scrollable grid.
- _____: Lays widgets out as a scrollable list.
- _____: Overlaps a widget on top of another.

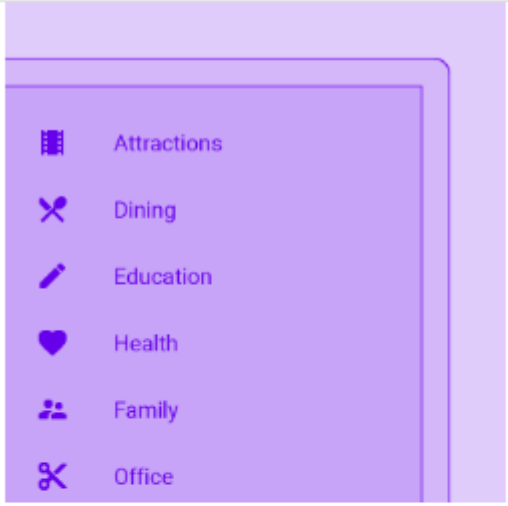
# Multi-child layout widgets

Layout a list of child widgets in the vertical direction.

Layout a list of child widgets in the horizontal direction.

.

# Multi-child layout widgets




A scrollable, linear list of widgets. ListView is the most commonly used scrolling widget. It displays its children one after another in the scroll direction....

This class is useful if you want to overlap several children in a simple way, for example having some text and an image, overlaid with...

A widget that uses the table layout algorithm for its children.

# Student Learning Outcomes

- Identify the layers of gesture system.
- Identify the different kinds of gestures.
- Create programs using various gestures.

# GestureDetector

- Flutter provides an excellent support for all type of gestures through its exclusive widget, GestureDetector.
- GestureDetector is a non-visual widget primarily used for detecting the user’s gesture.
- To identify a gesture targeted on a widget, the widget can be placed inside GestureDetector widget.
- GestureDetector will capture the gesture and dispatch multiple events based on the gesture.

# Gesture Detector

```
• body: Center(
•    child: GestureDetector(
•        onTap: () {
•            _showDialog(context);
•        },
•        child: Text( 'Hello World', )
•    )
• ),
```

```
/ user defined function void _showDialog(BuildContext context) {
  // flutter defined function
  showDialog(
    context: context, builder: (BuildContext context) {
      // return object of type Dialog
      return AlertDialog(
        title: new Text("Message"),
        content: new Text("Hello World"),
        actions: <Widget>[
          new FlatButton(
            child: new Text("Close"),
            onPressed: () {
              Navigator.of(context).pop();
            },
          ),
        ],
      );
    },
  );
```

# The gesture system in Flutter has two separate layers:

1. The first layer has raw pointer events that describe the location and movement of pointers (for example, touches, mice, and styli) across the screen.
2. The second layer has gestures that describe semantic actions that consist of one or more pointer movements.

# Gestures

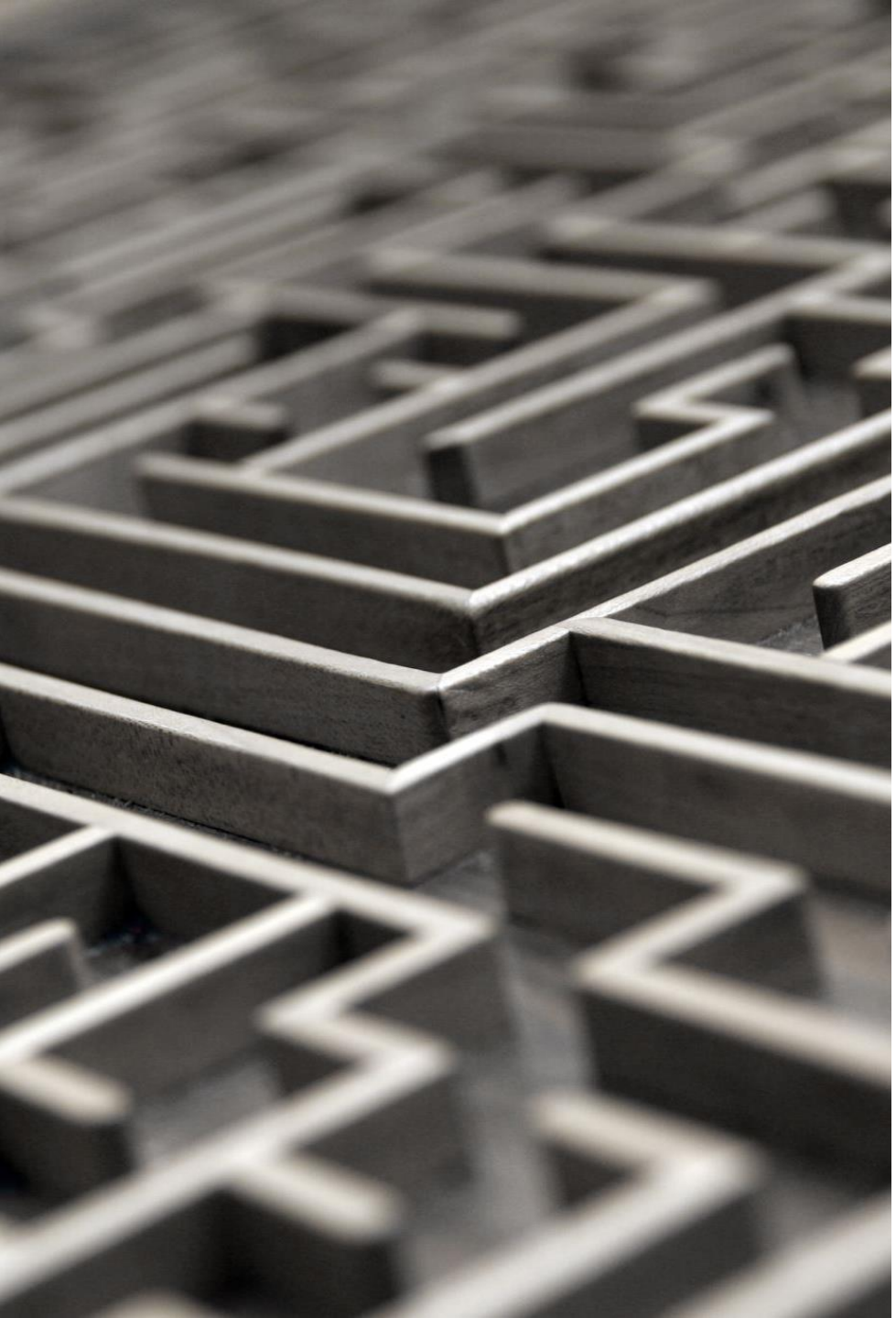
- Tap
- onTapDown
- A pointer that might cause a tap has contacted the screen at a particular location.
- onTapUp
- A pointer that will trigger a tap has stopped contacting the screen at a particular location.
- onTap
- The pointer that previously triggered the onTapDown has also triggered onTapUp which ends up causing a tap.
- onTapCancel
- The pointer that previously triggered the onTapDown will not end up causing a tap.
- Double tap

- onDoubleTap

# Gestures

Gestures represent Semantic actions (for example, tap, drag, and scale) that are recognized from multiple individual pointer events, potentially even multiple individual pointers.

Gestures can dispatch multiple events, corresponding to the lifecycle of the gesture (for example, drag start, drag update, and drag end):



# Gestures

- **Double tap**

- onDoubleTap
- The user has tapped the screen at the same location twice in quick succession.

- **Long press**

- onLongPress
- A pointer has remained in contact with the screen at the same location for a long period of time.

# Gestures

- Vertical drag

- **onVerticalDragStart**
- A pointer has contacted the screen and might begin to move vertically.
- **onVerticalDragUpdate**
- A pointer that is in contact with the screen and moving vertically has moved in the vertical direction.
- **onVerticalDragEnd**
- A pointer that was previously in contact with the screen and moving vertically is no longer in contact with the screen and was moving at a specific velocity when it stopped contacting the screen.

# Gestures

- Horizontal drag

- onHorizontalDragStart
- A pointer has contacted the screen and might begin to move horizontally.
- onHorizontalDragUpdate
- A pointer that is in contact with the screen and moving horizontally has moved in the horizontal direction.
- onHorizontalDragEnd
- A pointer that was previously in contact with the screen and moving horizontally is no longer in contact with the screen and was moving at a specific velocity when it stopped contacting the screen.

# Gestures

- Pan
- onPanStart
- A pointer has contacted the screen and might begin to move horizontally or vertically. This callback causes a crash if onHorizontalDragStart or onVerticalDragStart is set.
- onPanUpdate
- A pointer that is in contact with the screen and is moving in the vertical or horizontal direction. This callback causes a crash if onHorizontalDragUpdate or onVerticalDragUpdate is set.
- onPanEnd
- A pointer that was previously in contact with screen is no longer in contact with the screen and is moving at a specific velocity when it stopped contacting the screen. This callback causes a crash if onHorizontalDragEnd or onVerticalDragEnd is set.

# (9) GestureDetector (Widget of the Week) – YouTube

https://www.youtube.com/watch?v=WhVXkCFPmK4&t=12s

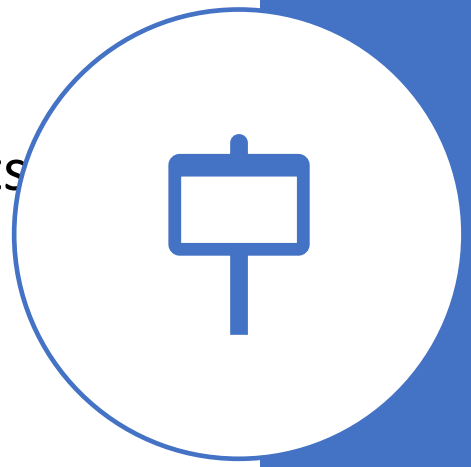
# Gestures

using Material Components, many of those widgets already respond to taps or gestures.

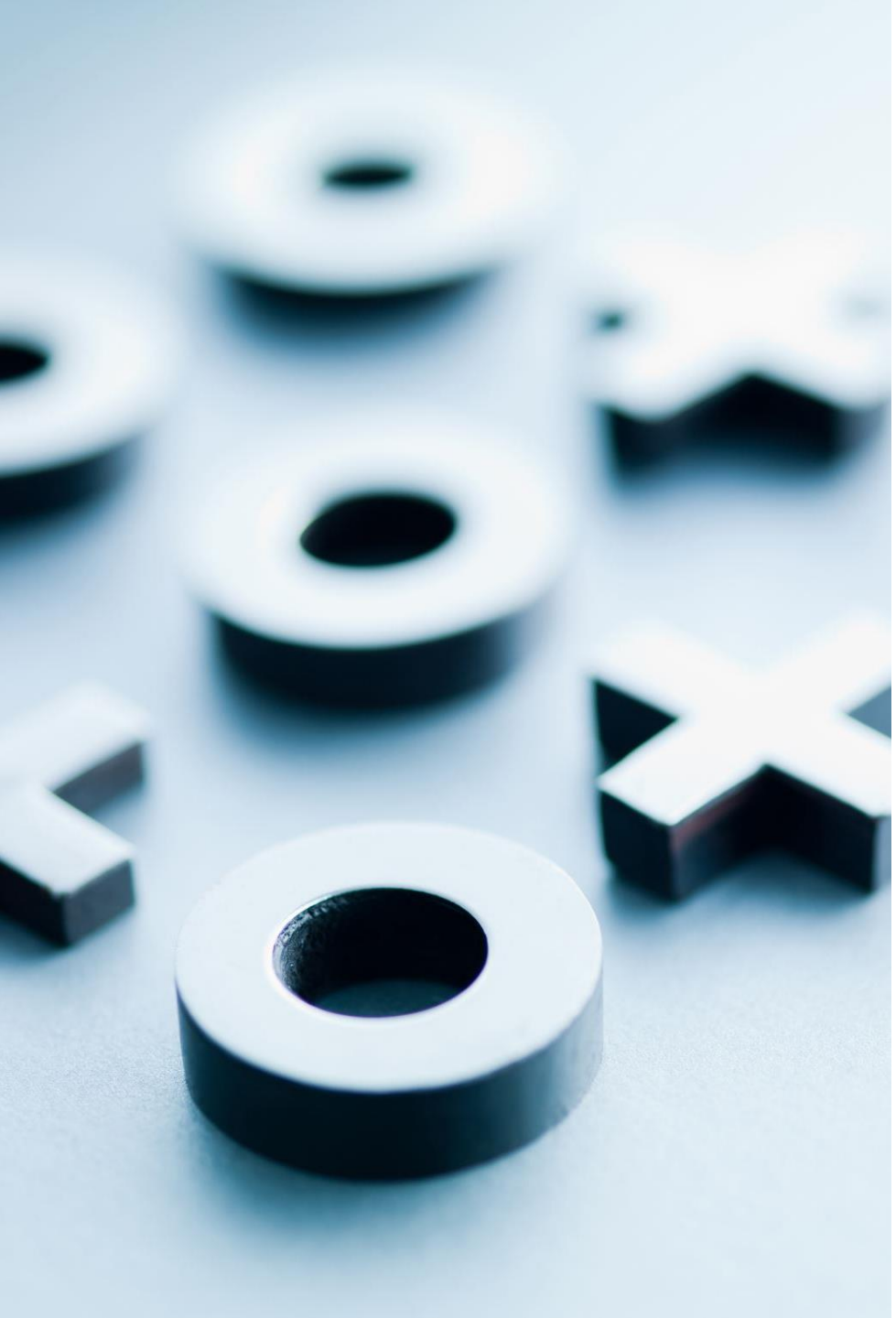
For example, IconButton and TextButton respond to presses (taps), and ListView responds to swipes to trigger scrolling.

# Gesture disambiguation

- At a given location on screen, there might be multiple gesture detectors.
- All of these gesture detectors listen to the stream of pointer events they flow past and attempt to recognize specific gestures.
- The GestureDetector widget decides which gestures to attempt to recognize based on which of its callbacks are non-null.



- When there is more than one gesture recognizer for a given pointer on the screen, the framework disambiguates which gesture the user intends by having each recognizer join the gesture arena.



# Gestures

- When there is more than one gesture recognizer for a given pointer on the screen, the framework disambiguates which gesture the user intends by having each recognizer join the gesture arena. The gesture arena determines which gesture wins using the following rules:

- At any time, a recognizer can declare defeat and leave the arena. If there’s only one recognizer left in the arena, that recognizer is the winner.

- At any time, a recognizer can declare victory, which causes it to win and all the remaining recognizers to lose.

# Pointers

- Pointers represent raw data about the user’s interaction with the device’s screen. There are four types of pointer events:

- PointerDownEvent
- The pointer has contacted the screen at a particular location.
- PointerMoveEvent
- The pointer has moved from one location on the screen to another.
- PointerUpEvent
- The pointer has stopped contacting the screen.
- PointerCancelEvent
- Input from this pointer is no longer directed towards this app.

# Pointers

- On pointer down, the framework does a hit test on your app to determine which widget exists at the location where the pointer contacted the screen.
- The pointer down event (and subsequent events for that pointer) are then dispatched to the innermost widget found by the hit test.
- From there, the events bubble up the tree and are dispatched to all the widgets on the path from the innermost widget to the root of the tree. There is no mechanism for canceling or stopping pointer events from being dispatched further.

- To listen to pointer events directly from the widgets layer, use a Listener widget.

# Preview

- Assessment – Next Week
- Presentation of Final Output – January 12,2023